

LECTURE 2

Estimation: Unbiasedness, Maximum Likelihood, and Confidence

STAT S4204 | Statistical Inference

Monday–Thursday, 13–16 July 2026

...the object of statistical methods is the reduction of data.

R. A. FISHER, “ON THE MATHEMATICAL FOUNDATIONS OF THEORETICAL STATISTICS” (1922)

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Overview. Estimation gets its own unit: the decision problem in which the action space *is* the parameter space and the loss is squared error. The week’s deck is an itinerary—most of its slides name what is then developed at the board—and these notes work each named item in full. Unbiasedness is defined in its general form (an estimand $g(\theta)$, an estimator $T(X)$) and defended as a way of choosing among admissible rules; the information bound of §1.9 is restated and then lifted to models with several parameters, where the price of nuisance parameters becomes visible. The unit’s centerpiece is the course’s first general-purpose *constructor* of estimators: maximum likelihood, whose estimate sits—approximately, when the information is high—at truth plus score over information, and therefore approximately attains the floor. Four classical samples make that exact or nearly so. Confidence sets then turn point estimates into precision statements via pivot statistics; conditional expectation returns as the best predictor under squared error; and when a

prior is on the table, the posterior mean is *proved* to minimize the Bayes risk—the theorem whose special case the trial met in 1.A.5—with the normal–normal, binomial–beta, and Poisson–gamma trios worked out. A closing pair of slides reads Bayes methods as shrinkage and penalization, and names the next horizon: sufficiency. HPS 2

2.1 Estimation as a Decision Problem

Unit 1 built a general machine—action space, loss function, method, risk (§ 1.5)—and then named its two workhorse special cases (§ 1.8). This week the first of them takes over. The deck’s opening slide re-fixes the frame in three bullets, and the first two objects below are those bullets; the third bullet—“squared error loss is a reasonable choice”—turns out to lean on the second in a way worth making explicit.

Concept 2.1.1

Estimation, resumed

Estimation is the class of “decision problems where the action space is the parameter space” (1.8.1): the act is a guess at the unknown, and reporting a when the truth is θ costs $(a - \theta)^2$ —“squared error loss is a reasonable choice.” Everything Unit 1 said about methods and risk applies verbatim; what is new this week is structure special to this action space: a floor on risk, a machine for building estimators, and a way to report precision.

Remark 2.1.2

“Close parameters are good guesses”

The deck’s middle bullet is a condition, not a throwaway: “parameterization satisfies: close parameters are good guesses.” Squared error charges by *distance in the parameter*, so it is a sensible price only when parameter distance tracks practical distance—when being off by 0.01 matters about equally everywhere in Θ . That is a property of the *parameterization*, not of nature: the same family of distributions can be labeled in ways that make squared error reasonable or misleading. The exponential sample of 2.5.4 will make this concrete—labeled by its mean, the model hands the obvious estimator a clean optimality; labeled by its rate, the “same” estimator is not even unbiased. Choosing the parameterization is part of posing the estimation problem.

Definition 2.1.3

Estimand and estimator

Let $\{P_\theta : \theta \in \Theta\}$ be a statistical model (1.3.1) and let X be a random element—a single observation, a vector, or an entire data set—whose distribution is P_θ for some unknown $\theta \in \Theta$. The *estimand* is a map $g : \Theta \rightarrow \mathbb{R}^k$, the quantity to be estimated; an *estimator* of $g(\theta)$ is a statistic

$$T = T(X),$$

a function of the data alone.¹

Definition 2.1.4

Unbiased estimator, general case

The estimator $T = T(X)$ is *unbiased* for $g(\theta)$ if

$$\mathbb{E}^\theta\{T(X)\} = g(\theta) \quad \text{for all } \theta \in \Theta;$$

¹Letters follow the deck’s typeset Definition 1: data X , estimator T , estimand g (the week-1 deck wrote data Y and rules δ ; the Fisher handout wrote $d(Y)$ —see 1.9.1). The deck also switches the parameter to a subscript, $\mathbb{E}_\theta$; these notes keep the superscript $\mathbb{E}^\theta$ of 1.3.1—same object, same fixed candidate truth.

equivalently, if its *bias* vanishes identically:

$$\text{Bias}^\theta(T) := \mathbb{E}^\theta\{T(X)\} - g(\theta) = 0 \quad \text{for all } \theta \in \Theta.$$

Remark 2.1.5

Reading the definition

The teeth are, as always, in “for all θ ”: the identity must hold at every candidate truth, not at a lucky one—this is the same quantifier that powered the differentiation trick of 1.9.10. The general g earns its keep at once: in the trial the estimand is $g(\theta_T, \theta_P) = \theta_T - \theta_P$, not the parameter itself, and 1.9.1 is recovered by taking g the identity. For $k > 1$ the display is read coordinate by coordinate. And the usual case discipline applies (1.1.7): the *estimator* T is a random variable; the *estimate* $t = T(x)$ is the number computed from your data.

Example 2.1.6

Unbiased or not: three classics

(a) *The sample mean.* For draws $X_1, \dots, X_n$ with common expectation μ , linearity gives $\mathbb{E}\{\bar{X}\} = \frac{1}{n} \sum_i \mathbb{E}\{X_i\} = \mu$: the average is unbiased for μ in *any* model with a mean—no shape assumptions used. (b) *The observed rate.* The arm total’s rate X/n is unbiased for a cure probability (1.A.3), and 1.A.6 showed it is, among unbiased estimators, unimprovable. (c) *The sample variance, and the $n - 1$ mystery.* The natural plug-in $\widehat{\sigma^2} = \frac{1}{n} \sum_i (X_i - \bar{X})^2$ is *biased*:

$$\mathbb{E}\left\{\frac{1}{n} \sum_i (X_i - \bar{X})^2\right\} = \frac{n-1}{n} \sigma^2 < \sigma^2,$$

because deviations are measured from $\bar{X}$, which has chased the data part of the way (the computation, every step, is §2.A.1). Dividing by $n - 1$ instead of n repairs the average exactly—that, and nothing deeper, is where the $n - 1$ comes from. The repaired version is unbiased; whether it is *better* is a risk question, not an unbiasedness question.

Concept 2.1.7

Why restrict to unbiased estimators?

The deck’s four bullets, in order. “*Need to have a way to choose from admissible estimators*”: admissibility (1.5.13) discards the dominated but leaves a crowd whose risk curves cross; restricting the contest is one way to get a winner. “*It makes sense*”: deadpan, but real—a rule that is systematically off at some θ is lying about that corner of the parameter space, and forbidding that is a defensible ground rule. “*The class will typically have a unique uniformly optimum*”: within the unbiased class, risk is variance (1.9.2), and in well-behaved models one estimator has the smallest variance at *every* θ simultaneously—a *uniformly* best member (the usual name, for later, is UMVU: uniformly minimum-variance unbiased). No such uniform winner exists among all rules (the crossing curves of 1.5.14); inside the class, the contest has a champion. “*Unbiased estimators in largish samples with smallish-dimensional parameters are usually just fine*”: the honest disclaimer—with much data and few parameters, the bias one could profitably trade away is small anyway. Unit 1’s counterpoint stands: with little data, a shaded (biased) estimator can beat every unbiased one over most of the parameter space (1.A.4). The restriction is a choice, and the deck presents it as one.

2.2 The Efficiency Bound, Revisited

“We did the efficiency bound last time,” the deck’s fifth slide begins, and compresses §1.9 into four bullets. The compression is worth reading closely—it contains the whole proof—and then the section does what the deck does next: it removes the assumption that θ is the only unknown.

Concept 2.2.1

The bound in one paragraph

The deck’s recap, unpacked. “Deviation of unbiased estimate from Score divided by information is uncorrelated with Score (because derivative of expectation is covariance with score)”: differentiating the unbiasedness identity showed $\text{Cov}^\theta(T, U) = 1$ (1.9.10, step (2)), and since $U/I(\theta)$ has $\text{Cov}(U/I, U) = I/I = 1$ as well, the residual $T - \theta - U/I(\theta)$ has covariance $1 - 1 = 0$ with the score: the projection geometry of 1.9.11. The variance of that projection, $U/I(\theta)$, is $1/I(\theta)$ —no larger than the variance of the unbiased estimator, which carries the residual too.² “And variance of score is the information...”: 1.9.8. And the conclusion, paraphrased (the original carries two typos, preserved in the transcript): the variance of an unbiased estimate is no less than the inverse information—the bound, 1.9.10.

Definition 2.2.2

Score vector and information matrix

For a model $f(x; \theta)$ with $\theta = (\theta_1, \dots, \theta_p)^\top \in \Theta \subseteq \mathbb{R}^p$ smooth in θ , the *score vector* is

$$U(\theta) = \left(\frac{\partial}{\partial \theta_1} \log f(X; \theta), \dots, \frac{\partial}{\partial \theta_p} \log f(X; \theta) \right)^\top,$$

and the *information matrix* is the $p \times p$ matrix

$$I(\theta) = \text{E}^\theta \{U(\theta) U(\theta)^\top\}, \quad I(\theta)_{jk} = \text{E}^\theta \{U_j(\theta) U_k(\theta)\}.$$

Each component score is the one-parameter score of 1.9.3 taken while the other coordinates hold still, so each has mean zero by 1.9.5 applied coordinatewise; the information matrix is therefore the covariance matrix of the score vector, and the argument of 1.9.7 gives, entry by entry, the curvature form $I(\theta)_{jk} = -\text{E}^\theta \{\partial^2 \log f / \partial \theta_j \partial \theta_k\}$. The bound now reads as follows; its proof is 1.9.10’s with one new ingredient, the matrix inverse.

Theorem 2.2.3

The information bound when there are other parameters

In a regular model with $I(\theta)$ invertible, let $\hat{\theta} = \hat{\theta}(X)$ be an unbiased estimator of the *first coordinate*: $\text{E}^\theta \{\hat{\theta}\} = \theta_1$ for all θ . Then

$$\text{Var}^\theta(\hat{\theta}) \geq e_1^\top I(\theta)^{-1} e_1 = (I(\theta)^{-1})_{11}, \quad e_1 = (1, 0, \dots, 0)^\top.$$

²As printed, the slide’s second bullet reads “So variance of score is less than variance of unbiased estimate”—a compression: the variance of the *score* is $I(\theta)$, typically enormous; the object meant is the variance of the *score divided by the information*, i.e. of the projection. The slide’s own first bullet has it right. (Recorded in the errata file.)

Proof 2.2.3

Proof of the Multiparameter Bound by Projecting onto the Span of the Score Components.

- (1) First, one covariance per coordinate. Differentiate the unbiasedness identity $\int \hat{\theta}(x)f(x;\theta) dx = \theta_1$ in each θ_j , passing the derivative under the integral and substituting $\partial f/\partial\theta_j = U_j f$ as in 1.9.10(2):

$$\mathbb{E}^\theta\{\hat{\theta} U_j(\theta)\} = \frac{\partial \theta_1}{\partial \theta_j} = \begin{cases} 1, & j = 1 \\ 0, & j \neq 1 \end{cases} \quad \text{i.e.} \quad \mathbb{E}^\theta\{\hat{\theta} U(\theta)\} = e_1.$$

Since each U_j has mean zero, the same holds with $\hat{\theta} - \theta_1$ in place of $\hat{\theta}$: the deviation covaries with the score components according to e_1 .

- (2) Next, the candidate projection: consider the scalar random variable $e_1^\top I(\theta)^{-1} U(\theta)$, a fixed linear combination of the score components. Its covariances with the components are

$$\mathbb{E}^\theta\{(e_1^\top I^{-1} U) U^\top\} = e_1^\top I^{-1} \mathbb{E}^\theta\{U U^\top\} = e_1^\top I^{-1} I = e_1^\top.$$

Identical to step (1)—so the difference is orthogonal to every score component:

$$\mathbb{E}^\theta\left[\left(\hat{\theta} - \theta_1 - e_1^\top I(\theta)^{-1} U(\theta)\right) U(\theta)\right] = \mathbf{0}.$$

This is the deck's displayed equation (its θ_0 is the true parameter value, the estimand being its first coordinate); “geometrically, we are projecting the unbiased estimator onto the span of the components of the score.”

- (3) The projection's variance, by squaring the linear combination:

$$\text{Var}^\theta(e_1^\top I^{-1} U) = e_1^\top I^{-1} \mathbb{E}^\theta\{U U^\top\} I^{-1} e_1 = e_1^\top I^{-1} I I^{-1} e_1 = e_1^\top I^{-1} e_1 = (I^{-1})_{11}.$$

- (4) Pythagoras finishes it: by step (2) the deviation splits into the projection plus an orthogonal residual, so variances add as in 1.9.11:

$$\text{Var}^\theta(\hat{\theta}) = (I^{-1})_{11} + \text{Var}^\theta(\hat{\theta} - \theta_1 - e_1^\top I^{-1} U) \geq (I^{-1})_{11}.$$

■

Uses 1.9.5, 1.9.10, 2.2.2.

Remark 2.2.4

The geometry, upgraded

Nothing changed but the dimension of the stage. In 1.9.11 the score spanned a *line* and the bound was the squared length of a projection onto it; here the p score components span a p -dimensional

subspace, the projection of $\hat{\theta} - \theta_1$ onto that subspace is $e_1^\top I^{-1}U$, and the right triangle of 1.9.12 is intact—hypotenuse the deviation, base the projection, and the bound again the base’s squared length.

Remark 2.2.5

Nuisance parameters cost information

Why $(I^{-1})_{11}$ and not $1/I_{11}$? Because they differ, and the difference is the price of company:

$$(I^{-1})_{11} \geq \frac{1}{I_{11}}, \quad \text{with equality if and only if } I_{1j} = 0 \text{ for all } j \neq 1$$

(the 2×2 case, with every step, is §2.A.2). If the other coordinates were *known*, the model would be one-parameter and the floor would be $1/I_{11}$; having to carry them raises the floor unless their scores are uncorrelated with the first’s—the *orthogonal* case. The normal sample of 2.6.3 lands exactly there: its information matrix is diagonal, so not knowing the variance costs the mean’s estimator nothing. When the off-diagonals do not vanish, estimating your parameter of interest is genuinely harder because you must estimate its neighbors’ influence too.

2.3 Two Information Computations

The deck’s seventh slide names two models and then asks, flatly, “What is information?” The two computations first; the question, with its three answers, after.

Example 2.3.1

Binomial

One arm of the trial, revisited with the new vocabulary: $X \sim \text{Binomial}(n, \theta)$ (the total of n iid Bernoulli variables). Its information was computed, both ways and with every step, in 1.A.6:

$$I_n(\theta) = \frac{n}{\theta(1-\theta)}, \quad \frac{1}{I_n(\theta)} = \frac{\theta(1-\theta)}{n} = \text{Var}^\theta(X/n):$$

the observed rate attains the floor. Note where the floor is *highest*: $\theta(1-\theta)$ peaks at $\theta = \frac{1}{2}$, so the middle of the parameter space is the noisiest place to estimate a proportion—and the information, its reciprocal (times n), is smallest there. A coin near fair is the hardest coin to pin down.

Example 2.3.2

Multivariate normal with known covariance

Let $Y \sim N_p(\theta, \Sigma)$ with Σ known and invertible: the estimand is the mean vector itself. The log-density is

$$\log f(y; \theta) = -\frac{1}{2} (y - \theta)^\top \Sigma^{-1} (y - \theta) - \frac{1}{2} \log((2\pi)^p |\Sigma|),$$

and differentiating in θ kills the constant and unfolds the quadratic:

$$U(\theta) = \Sigma^{-1}(Y - \theta), \quad I(\theta) = \text{E}^\theta \{ \Sigma^{-1}(Y - \theta)(Y - \theta)^\top \Sigma^{-1} \} = \Sigma^{-1} \Sigma \Sigma^{-1} = \Sigma^{-1}.$$

The information is the *precision matrix*—the covariance’s inverse—and it does not depend on θ . For an iid sample of size n the information is $n\Sigma^{-1}$ (1.9.9, coordinatewise), the floor for the first coordinate is $((n\Sigma^{-1})^{-1})_{11} = \Sigma_{11}/n$, and $\bar{X}_1$ —with variance exactly Σ_{11}/n —attains it.

Remark 2.3.3

“What is information?”

The deck’s question, and the three answers the course now owns. *Algebraically*: the variance (matrix) of the score—how much the data’s argument about θ swings from sample to sample (1.9.8). *Geometrically*: the expected curvature of the log-likelihood at the truth—how sharply the model distinguishes θ from its neighbors (1.9.7). *Operationally*: an inverse variance—the best precision any unbiased estimator can achieve, attained in the two examples above. And it is additive: n independent observations carry n times the information of one (1.9.9)—the sense in which evidence accumulates.

2.4 Maximum Likelihood

Everything so far judges estimators after someone proposes one; the floor of 2.2.3 says how good a proposal could possibly be, and stays silent about where proposals come from. The deck’s eighth slide supplies the factory. The idea is a reversal: the density $f(x; \theta)$ was built to be read as a function of x at a fixed θ —probabilities of data under a hypothesis. Read it the other way.

Definition 2.4.1

The likelihood

Given observed data X , the *likelihood function* of the model $f(x; \theta)$ is

$$\text{lik}(\theta) = f(X; \theta), \quad \theta \in \Theta,$$

the joint density (or pmf) of the data, evaluated at the data, read as a function of the parameter. Its logarithm $\ell(\theta) = \log \text{lik}(\theta)$ is the *log-likelihood*.

Remark 2.4.2

Reading the likelihood

$\text{lik}(\theta)$ answers, for each candidate truth: *how probable was what I actually saw, if this candidate were the truth?* It ranks explanations. It is *not* a probability distribution over θ —nothing says $\int \text{lik}(\theta) d\theta = 1$, and no randomness attaches to θ here³—only its *ratios* across candidate θ carry meaning, which is why multiplicative constants (in θ) are routinely dropped. The score of 1.9.3 is exactly the log-likelihood’s slope: the two sections have been talking about the same curve all along.

Definition 2.4.3

Maximum likelihood estimator

The *maximum likelihood estimator* (MLE) is the parameter value at which the likelihood—equivalently the log-likelihood—is largest:

$$\hat{\theta} = \hat{\theta}(X) = \arg \max_{\theta \in \Theta} \text{lik}(\theta) = \arg \max_{\theta \in \Theta} \ell(\theta).$$

³The distinction is worth a sentence, because the word “likely” invites the slide. “The probability of the data given θ ” is a statement about *data*; the likelihood re-reads that number as a score for θ , the data now frozen. A statement like “ θ is probably near 0.4” needs θ to be random—which is Bayesian territory, where § 2.9’s posterior, prior $\times$ likelihood normalized, makes it legitimate.

Remark 2.4.4*“The score!”*

The deck’s sub-bullet under “The MLE” is two words and an exclamation point, and it is the right reaction. If the maximum sits in the interior of Θ and ℓ is smooth, the first-order condition is

$$0 = \ell'(\hat{\theta}) = U(\hat{\theta}) :$$

the MLE is the parameter value at which the score vanishes—where the data stop arguing for a nudge in either direction (1.9.4). The object built in §1.9 to measure estimators turns out to be the equation that constructs one.

Concept 2.4.5*Truth plus score over information*

The deck: “The MLE differs from parameter by the ratio of score to information (approximately with high information).” Here is the calculation behind the sentence. Expand the score equation $0 = \ell'(\hat{\theta})$ in a Taylor series about the true value θ :

$$0 = \ell'(\hat{\theta}) \approx \ell'(\theta) + (\hat{\theta} - \theta) \ell''(\theta),$$

keeping the linear term. Solve for the deviation:

$$\hat{\theta} - \theta \approx \frac{\ell'(\theta)}{-\ell''(\theta)} = \frac{U(\theta)}{-\ell''(\theta)} \approx \frac{U(\theta)}{I(\theta)},$$

the last step replacing the realized curvature $-\ell''(\theta)$ by its expectation $I(\theta)$ (1.9.7)—a substitution that is good when the information is high, because the curvature, a sum of n independent contributions, then concentrates near its mean. Both approximations sharpen as $I(\theta)$ grows; neither is a theorem as stated, and the deck’s parenthetical—“approximately with high information”—is the honest label. The notes will treat it as the deck does: a heuristic with excellent credentials, exact in the cleanest models (2.5.1) and provable-with-care in wide generality (a story for a later course).

Remark 2.4.6*What the heuristic buys*

Take the approximation $\hat{\theta} - \theta \approx U(\theta)/I(\theta)$ seriously and compute with it. The right side has expectation $E^\theta\{U\}/I = 0$ (1.9.5): the MLE is *approximately unbiased*. Its variance is $\text{Var}^\theta(U)/I(\theta)^2 = I(\theta)/I(\theta)^2 = 1/I(\theta)$ (1.9.8): the MLE *approximately attains the information bound*. That is the week’s arc in one line: the floor of §1.9 is not just a benchmark—maximum likelihood is a general-purpose recipe that (approximately, eventually) reaches it. And the expression U/I is an old friend: it is exactly the quantity whose *exact* equality with $\hat{\theta} - \theta$ characterized the models where the bound is hit on the nose (1.9.11, 1.9.13).



Figure 2.4.7. LIKELIHOOD, SCORE, AND INFORMATION IN ONE PICTURE. Two log-likelihoods with the same maximizer $\hat{\theta}$ (dashed drop). The slope of the curve is the score, zero at the peak (2.4.4); at a candidate θ left of the peak the slope is positive—the data argue for moving right. The curvature is the information’s realized version: on the left the curve is sharply bent ($-\ell''$ large), so parameter values away from $\hat{\theta}$ are much worse explanations and the peak deserves trust; on the right the same peak sits on a nearly flat ridge ($-\ell''$ small)—many candidates explain the data almost equally well, and the estimate, while the same number, means less. The information bound $1/I$ of 1.9.10 is this picture’s moral quantified.

2.5 Four Samples, Four Maximum-Likelihood Estimators

The deck’s ninth slide is four one-word prompts—“Normal, Bernoulli, Exponential, Poisson”—and two questions: “What is the MLE,” and “How does it compare to the information inequality?” As with Unit 1’s closing checklist, the notes work what the deck names. Each example runs the same drill: likelihood, log, score, MLE, information, and the comparison. Throughout, $X_1, \dots, X_n$ are iid and $\bar{X}$ is their average.

Example 2.5.1

Normal, known variance

$X_i \sim N(\mu, \sigma^2)$ with σ^2 known. The score was computed in 1.9.13: $U(\mu) = n(\bar{X} - \mu)/\sigma^2$, which vanishes exactly at

$$\hat{\mu} = \bar{X}.$$

(The log-likelihood is a downward parabola in μ , so the critical point is the global maximum.) Information $I_n(\mu) = n/\sigma^2$; floor σ^2/n ; and $\text{Var}^\mu(\bar{X}) = \sigma^2/n$: the MLE is unbiased and attains the bound *exactly, at every n* —the degenerate-triangle model of 1.9.13, where $\hat{\mu} - \mu = U(\mu)/I(\mu)$ holds with no approximation at all.

Example 2.5.2

Bernoulli

$X_i \sim \text{Bernoulli}(\theta)$, $0 < \theta < 1$, with arm total $X = \sum_i X_i$. The log-likelihood and score, from 1.A.6(1)–(2) summed over the sample:

$$\ell(\theta) = X \log \theta + (n - X) \log(1 - \theta), \quad U(\theta) = \frac{X}{\theta} - \frac{n - X}{1 - \theta} = \frac{X - n\theta}{\theta(1 - \theta)}.$$

The score vanishes exactly when $X = n\theta$, i.e. at

$$\hat{\theta} = \frac{X}{n} = \bar{X}$$

(and $\ell'' < 0$ throughout, so this is the maximum; for the boundary samples $X = 0$ and $X = n$ the likelihood is monotone, and its maximum sits at the same X/n , now an endpoint). The observed rate—Unit 1's estimator of choice—is the MLE. Information $I_n(\theta) = n/(\theta(1-\theta))$ (1.A.6); floor $\theta(1-\theta)/n$; variance of the rate: the same (1.A.3). Attained exactly.

Example 2.5.3

Poisson

$X_i \sim \text{Poisson}(\lambda)$, $\lambda > 0$: pmf $f(x; \lambda) = e^{-\lambda} \lambda^x / x!$, so

$$\ell(\lambda) = -n\lambda + \left(\sum_i X_i \right) \log \lambda - \sum_i \log(X_i!), \quad U(\lambda) = -n + \frac{\sum_i X_i}{\lambda} = \frac{\sum_i X_i - n\lambda}{\lambda}.$$

For samples with $\sum_i X_i > 0$ the score vanishes at $\hat{\lambda} = \bar{X}$, and the second derivative $-\sum_i X_i / \lambda^2 < 0$ confirms a maximum (an all-zero sample makes the likelihood $e^{-n\lambda}$ monotone decreasing, driving the maximizer to the boundary, where $\bar{X} = 0$ still names it). Information, by the curvature form (1.9.7): $I_n(\lambda) = E\{\sum_i X_i\} / \lambda^2 = n\lambda / \lambda^2 = n/\lambda$. Floor λ/n ; and since a Poisson variable has variance equal to its mean, $\text{Var}(\bar{X}) = \lambda/n$. Attained exactly—and indeed $U(\lambda)/I_n(\lambda) = \frac{\sum_i X_i - n\lambda}{\lambda} \cdot \frac{\lambda}{n} = \bar{X} - \lambda$, the equality case once more.

Example 2.5.4

Exponential: one sample, two parameterizations

Exponential variables can be labeled by their *mean* μ (density $f(x; \mu) = \mu^{-1} e^{-x/\mu}$, $x > 0$) or by their *rate* $\lambda = 1/\mu$ (density $\lambda e^{-\lambda x}$). The choice, innocuous-looking, decides how estimation goes—the promise of 2.1.2 kept.

Mean parameterization.

$$\ell(\mu) = -n \log \mu - \frac{n\bar{X}}{\mu}, \quad U(\mu) = -\frac{n}{\mu} + \frac{n\bar{X}}{\mu^2} = \frac{n(\bar{X} - \mu)}{\mu^2},$$

so $\hat{\mu} = \bar{X}$: unbiased (linearity). Information: differentiating, $\ell''(\mu) = n/\mu^2 - 2n\bar{X}/\mu^3$, so $I_n(\mu) = -E\{\ell''(\mu)\} = -n/\mu^2 + 2n\mu/\mu^3 = n/\mu^2$; the floor is μ^2/n ; and since an exponential's variance is its squared mean, $\text{Var}(\bar{X}) = \mu^2/n$. Attained exactly ($U/I_n = \bar{X} - \mu$ once again).

Rate parameterization.

$$\ell(\lambda) = n \log \lambda - \lambda n \bar{X}, \quad U(\lambda) = \frac{n}{\lambda} - n \bar{X},$$

so $\hat{\lambda} = 1/\bar{X}$ —by invariance the natural counterpart, but *not* unbiased:

$$E^\lambda\{\hat{\lambda}\} = \frac{n\lambda}{n-1} > \lambda \quad (n \geq 2),$$

computed through the Gamma distribution of the sample total in §2.A.3. The bound of 1.9.10 does not even apply to it as stated; and the repaired, unbiased version $(n-1)/(n\bar{X})$ has variance $\lambda^2/(n-2)$

(§2.A.3)—strictly *above* the floor λ^2/n , at every $n \geq 3$. Same data, same principle; the parameterization alone separates “exactly optimal” from “biased, and above the floor even after repair.”

Remark 2.5.5

The pattern, and its limits

The deck’s second question—“How does it compare to the information inequality?”—now has four answers:

model	MLE	I_n	floor $1/I_n$	bound attained?
Normal mean (σ^2 known)	$\bar{X}$	n/σ^2	σ^2/n	yes—exactly
Bernoulli θ	X/n	$n/(\theta(1-\theta))$	$\theta(1-\theta)/n$	yes—exactly
Poisson λ	$\bar{X}$	n/λ	λ/n	yes—exactly
Exponential mean μ	$\bar{X}$	n/μ^2	μ^2/n	yes—exactly
Exponential rate λ	$1/\bar{X}$	n/λ^2	λ^2/n	no—biased; repaired, still above

The run of “exactly” is not the general situation—it is the luck of these particular families under these particular labelings, the models where $\theta + U(\theta)/I(\theta)$ happens to assemble into a statistic (1.9.11)—here, each time, into $\bar{X}$ itself. The exponential-rate row shows how thin the luck is: a smooth relabeling of the *same* model breaks exactness. What survives relabeling is 2.4.6’s asymptotic version—approximately unbiased, approximately at the floor, when the information is high—and that, not exactness, is maximum likelihood’s true general credential.

2.6 Several Parameters at Once

Real models rarely have a single unknown. The deck’s tenth slide asks for the multi-parameter version of 2.4.5 and of its variance consequence, then names two examples; 2.2.2’s score vector and information matrix are already on the table.

Concept 2.6.1

Taylor series expansion for the MLE

For $\theta \in \Theta \subseteq \mathbb{R}^p$, the MLE solves the p score equations $U(\hat{\theta}) = \mathbf{0}$. Expanding each coordinate of U about the truth, to first order,

$$\mathbf{0} = U(\hat{\theta}) \approx U(\theta) + \nabla U(\theta)(\hat{\theta} - \theta),$$

where $\nabla U(\theta)$ is the $p \times p$ matrix of second partials of ℓ . Replacing that realized curvature matrix by its expectation $-I(\theta)$ (2.2.2) and solving,

$$\hat{\theta} - \theta \approx I(\theta)^{-1} U(\theta) :$$

truth plus *matrix* inverse of information times score—2.4.5 with the division promoted to a solve. The same two approximations, with the same “high information” license.

Concept 2.6.2

Covariance matrix of the multivariate MLE

Compute with the approximation. The score vector has mean $\mathbf{0}$ and covariance matrix $I(\theta)$ (2.2.2), so $\hat{\theta}$ is approximately unbiased and

$$\text{Cov}(\hat{\theta}) \approx I^{-1} \text{E}\{UU^\top\} I^{-1} = I^{-1} I I^{-1} = I(\theta)^{-1} :$$

the full inverse information matrix. In particular $\text{Var}(\hat{\theta}_1) \approx (I^{-1})_{11}$ —each coordinate of the MLE approximately attains the several-parameter floor of 2.2.3, nuisance tax included.

Example 2.6.3

Normal sample, expectation and variance both unknown

$X_i \sim N(\mu, \sigma^2)$, both parameters unknown: $\theta = (\mu, \sigma^2)$. The log-likelihood is

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2,$$

with the two score components

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_i (X_i - \mu), \quad \frac{\partial \ell}{\partial (\sigma^2)} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_i (X_i - \mu)^2.$$

Setting both to zero: the first gives $\hat{\mu} = \bar{X}$ regardless of σ^2 ; substituting into the second and solving, $\frac{n}{2\sigma^2} = \frac{1}{2\sigma^4} \sum_i (X_i - \bar{X})^2$, so

$$\hat{\mu} = \bar{X}, \quad \widehat{\sigma^2} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Note which variance estimator maximum likelihood picked: the *biased* $1/n$ version of 2.1.6(c)—the two principles of this unit genuinely disagree here, by the factor $(n-1)/n$. For the information matrix, differentiate the scores once more and take expectations (using $\text{E}\{X_i - \mu\} = 0$ and $\text{E}\{(X_i - \mu)^2\} = \sigma^2$):

$$-\text{E} \left[\frac{\partial^2 \ell}{\partial \mu^2} \right] = \frac{n}{\sigma^2}, \quad -\text{E} \left[\frac{\partial^2 \ell}{\partial \mu \partial (\sigma^2)} \right] = \text{E} \left[\frac{\sum_i (X_i - \mu)}{\sigma^4} \right] = 0, \quad -\text{E} \left[\frac{\partial^2 \ell}{\partial (\sigma^2)^2} \right] = -\frac{n}{2\sigma^4} + \frac{n\sigma^2}{\sigma^6} = \frac{n}{2\sigma^4},$$

so

$$I(\mu, \sigma^2) = \begin{pmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{pmatrix}, \quad I(\mu, \sigma^2)^{-1} = \begin{pmatrix} \frac{\sigma^2}{n} & 0 \\ 0 & \frac{2\sigma^4}{n} \end{pmatrix}.$$

The matrix is *diagonal*: the two scores are uncorrelated, the parameters are orthogonal, and $(I^{-1})_{11} = \sigma^2/n = 1/I_{11}$ —by 2.2.5, not knowing the variance costs the mean's estimator nothing (to this order). The floor for $\hat{\mu}$ is the same σ^2/n as when σ^2 was known, and $\bar{X}$ still attains it.

Example 2.6.4

Multivariate normal with known covariance, again

For an iid sample from $N_p(\theta, \Sigma)$, 2.3.2 gives $I_n(\theta) = n\Sigma^{-1}$; the MLE solves $\sum_i \Sigma^{-1}(X_i - \theta) = \mathbf{0}$, i.e. $\hat{\theta} = \bar{X}$, whose covariance matrix is exactly $\Sigma/n = (n\Sigma^{-1})^{-1} = I_n(\theta)^{-1}$. Here 2.6.2's approximation is an identity—the multivariate analogue of the degenerate triangle.

2.7 Confidence Sets

An estimate alone is a bare number: $\hat{\theta} = 0.31$ does not say whether the data would have supported 0.30 and 0.32 almost equally (low information) or ruled them out (high). The deck’s eleventh slide asks for the definition and for its reading “as a measure of precision”; the twelfth builds the standard construction; the thirteenth, a title with no body, names a duality the testing unit will cash in.

Definition 2.7.1

Confidence set

A *confidence set* of level $1 - \alpha$ for the parameter θ is a set $S(X)$, computed from the data alone, such that

$$P^\theta\{\theta \in S(X)\} \geq 1 - \alpha \quad \text{for all } \theta \in \Theta.$$

When $S(X)$ is an interval it is called a *confidence interval*.

Remark 2.7.2

Reading “confidence”

Look hard at the display: the random object inside the braces is $S(X)$, *not* θ . The parameter sits still; the *set* jumps around with the data; the guarantee—at every candidate truth, and before the data arrive—is that the procedure’s set captures the truth with probability at least $1 - \alpha$. It is a statement about the *method*, in exactly 1.5.6’s sense of a plan graded before the data arrive.⁴ And the deck’s phrase “as a measure of precision” is the point of reporting the whole set: its *size* is the resolution of the study. 2.7.5 will make that literal—the width shrinks exactly like the square root of the inverse information.

Definition 2.7.3

Pivot statistic

A *pivot* is a function $T(X, \theta)$ of the data *and* the parameter whose distribution under P^θ is the same known distribution for every $\theta \in \Theta$.

Remark 2.7.4

From pivot to confidence set

A pivot converts distributional knowledge into coverage. Choose a region A with $P\{T(X, \theta) \in A\} = 1 - \alpha$ —possible because the distribution of the pivot is known and free of θ —and *invert*: define

$$S(X) = \{\theta : T(X, \theta) \in A\}.$$

Then for every θ ,

$$P^\theta\{\theta \in S(X)\} = P^\theta\{T(X, \theta) \in A\} = 1 - \alpha :$$

the two events are the *same* event, described from the two ends of the (data, parameter) pair. All the standard intervals are this one identity wearing different pivots.

⁴Hence the classical caution: after computing, say, the interval $[0.22, 0.40]$, the statement “ θ is in $[0.22, 0.40]$ with probability 0.95” is not licensed—nothing random remains in it. The honest post-data sentence is operational: the interval was produced by a procedure that traps the truth 95% of the time. A probability distribution *for* θ *itself* requires a prior, and then § 2.9’s posterior supplies genuine probability statements of just that kind.

Example 2.7.5

The z-interval: normal mean, known variance

$X_i \sim N(\mu, \sigma^2)$, σ^2 known. Since $\bar{X} \sim N(\mu, \sigma^2/n)$, standardizing gives

$$Z = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim N(0, 1) \quad \text{under } P^\mu, \text{ for every } \mu :$$

a pivot. Take $A = [-z_{\alpha/2}, z_{\alpha/2}]$, the central interval catching probability $1 - \alpha$ of the standard normal, and invert the two inequalities in μ :

$$-z_{\alpha/2} \leq \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \leq z_{\alpha/2} \iff \mu \in \left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right].$$

The width is $2z_{\alpha/2} \sigma / \sqrt{n} = 2z_{\alpha/2} \sqrt{1/I_n(\mu)}$: the information of §1.9 is not only the variance floor but the *ruler* for the precision report. Quadrupling the data halves the width.

Example 2.7.6

The t statistic: normal mean, unknown variance

With σ^2 unknown, the z pivot is not a statistic-and-parameter pair we can use— σ is not known. Replace it by the sample's own scale, $s^2 = \frac{1}{n-1} \sum_i (X_i - \bar{X})^2$ (the unbiased version of 2.1.6(c)). The resulting ratio

$$T = \frac{\sqrt{n}(\bar{X} - \mu)}{s}$$

is again a pivot: its distribution—Student's t with $n - 1$ degrees of freedom—is the same for every (μ, σ^2) (the classical distribution theory of the normal sample is taken from the text here rather than re-derived, HPS 2). Inverting exactly as before,

$$\mu \in \left[\bar{X} - t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}, \bar{X} + t_{n-1, \alpha/2} \frac{s}{\sqrt{n}} \right],$$

with $t_{n-1, \alpha/2} > z_{\alpha/2}$: the interval is systematically wider than the known- σ interval. That widening is 2.2.5's nuisance tax, paid in quantiles—not knowing σ^2 costs nothing in the information matrix (which is diagonal, 2.6.3) but does cost in having to estimate the ruler itself; the surcharge $t_{n-1, \alpha/2} - z_{\alpha/2}$ fades as n grows.

Example 2.7.7

Confidence sets for ratios

The deck's last worked prompt, and the reason it says confidence *sets*. Two independent sample means estimate μ_1 and μ_2 , say $\bar{X} \sim N(\mu_1, \sigma^2/n)$ and $\bar{Y} \sim N(\mu_2, \sigma^2/m)$ with σ^2 known, and the estimand is the ratio $\rho = \mu_1/\mu_2$, with $\mu_2 \neq 0$ so that the ratio is defined (a treatment-to-control ratio, a slope through the origin). For each candidate ρ , under any truth with $\mu_1 = \rho \mu_2$ the combination $\bar{X} - \rho \bar{Y}$ is normal with mean zero and known variance $\sigma^2(1/n + \rho^2/m)$, so

$$Z(\rho) = \frac{\bar{X} - \rho \bar{Y}}{\sigma \sqrt{1/n + \rho^2/m}} \sim N(0, 1),$$

and the inversion recipe of 2.7.4 yields

$$S(X, Y) = \left\{ \rho : (\bar{X} - \rho \bar{Y})^2 \leq z_{\alpha/2}^2 \sigma^2 \left(\frac{1}{n} + \frac{\rho^2}{m} \right) \right\}.$$

The condition is a *quadratic* inequality in ρ (Fieller’s construction), and the shape of its solution set depends on the leading coefficient $\bar{Y}^2 - z_{\alpha/2}^2 \sigma^2 / m$: an interval when $\bar{Y}$ is convincingly nonzero; two half-lines, or all of $\mathbb{R}$, when it is not (§2.A.5 runs the cases). The strange shapes are honesty, not pathology: when the denominator’s sign is in doubt, the data genuinely cannot bound the ratio, and a procedure with guaranteed coverage must say so.⁵

Concept 2.7.8

Duality with families of point-null tests

The deck’s thirteenth slide is a bare title: “Duality between families of point null tests and confidence sets.” The one-sentence preview: deciding whether each candidate value θ_0 is *plausible* (a test of the point null $\theta = \theta_0$, in the sense of 1.8.2) and collecting the plausible ones is the same act as reporting a confidence set— $\theta_0 \in S(X)$ exactly when the test of θ_0 accepts. 2.7.7 was secretly this: a family of z -tests, one per candidate ρ , inverted. The testing unit will make the correspondence precise and run it in both directions.

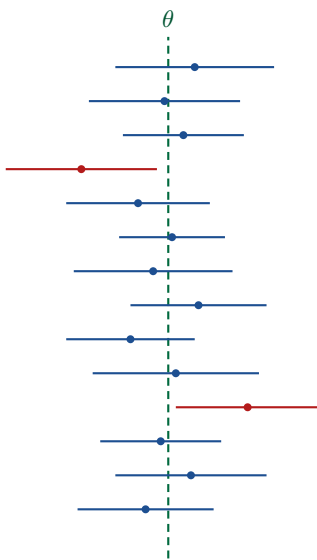


Figure 2.7.9. COVERAGE IS A PROPERTY OF THE PROCEDURE. Fourteen replications of the same experiment, each yielding an interval (dot at the estimate). The truth θ (dashed green) never moves; the intervals do. Twelve of the fourteen catch θ ; two (red) miss. “95% confidence” is a statement about the long-run frequency of catches over replications—not about any single computed interval, which either contains θ or does not (2.7.2).

2.8 Prediction, and the Return of Conditional Expectation

“More probability review,” the fourteenth slide announces, modestly. Its two questions are the engine of everything after: “How to minimize the expected prediction error under squared error

⁵An interval-valued report would have to fail its coverage promise for some (μ_1, μ_2) ; the exotic sets are the price of the “for all θ ” in 2.7.1.

loss?”—and the same again “when you have a predictor?” The answers are two small theorems about squared error, and the second is conditional expectation’s second star turn (after 1.4.16).

Theorem 2.8.1

The best constant predictor is the mean

Let Y have $E\{Y^2\} < \infty$. Among all constants c ,

$$E\{(Y - c)^2\} = \text{Var}(Y) + (E\{Y\} - c)^2$$

is minimized uniquely at $c = E\{Y\}$, with minimum value $\text{Var}(Y)$.

Proof 2.8.1

Proof of the Best Constant by Adding and Subtracting the Mean.

(1) Route the error through the mean: $Y - c = (Y - E\{Y\}) + (E\{Y\} - c)$.

(2) Square the two-term sum:

$$(Y - c)^2 = (Y - E\{Y\})^2 + 2(Y - E\{Y\})(E\{Y\} - c) + (E\{Y\} - c)^2.$$

(3) Take expectations term by term. The middle term dies— $E\{Y\} - c$ is a constant and deviations about the mean average to zero—leaving the display: $E\{(Y - c)^2\} = \text{Var}(Y) + (E\{Y\} - c)^2$.

(4) The first term does not involve c ; the second is a square, zero exactly at $c = E\{Y\}$ and positive elsewhere. ■

Uses 1.A.1.

Theorem 2.8.2

The best predictor using X is the conditional expectation

Let (X, Y) be jointly distributed with $E\{Y^2\} < \infty$. Among all functions h of the predictor X ,

$$E\{(Y - h(X))^2\}$$

is minimized by

$$h(X) = E\{Y \mid X\}.$$

Proof 2.8.2

Proof of the Best Predictor by Conditioning, Slice by Slice.

(1) Iterate the expectation through the predictor (1.4.16):

$$E\{(Y - h(X))^2\} = E\left[E\{(Y - h(X))^2 \mid X\}\right].$$

- (2) Fix a slice $X = x$. Inside it, $h(x)$ is a *constant*, and the conditional distribution of Y given $X = x$ is a genuine probability distribution (1.4.14)—so 2.8.1 applies verbatim *within the slice*:

$$\mathbb{E}\{(Y - h(x))^2 \mid X = x\} \geq \text{Var}(Y \mid X = x),$$

with equality exactly at $h(x) = \mathbb{E}\{Y \mid X = x\}$.

- (3) The choice $h(x) = \mathbb{E}\{Y \mid X = x\}$ therefore minimizes the inner expectation at *every* x simultaneously; averaging over the slices (step (1)) preserves the domination, so no other h can do better overall.

■

Uses 1.4.14, 1.4.16, 2.8.1.

Remark 2.8.3

The regression function

The minimizer $x \mapsto \mathbb{E}\{Y \mid X = x\}$ has a name—the *regression function* of Y on X —and the two theorems assign the season’s roles: with no predictor in hand, report the mean; with one, report the conditional mean. This is *prediction* (the target Y is a random quantity not yet seen), a cousin of estimation (the target $g(\theta)$ is a fixed unknown); the machinery of the regression weeks will be theorem 2.8.2 worked into linear form. More immediately: the deck placed this “review” exactly where it did because the Bayes section next door is 2.8.2 in disguise.

2.9 If You Have a Prior: the Posterior Mean

“What if you have a prior?” Unit 1 built the Bayesian scaffolding—prior, Bayes risk, Bayes principle (§ 1.6)—and, in the trial, found by direct computation that the Bayes rule was a posterior mean (1.A.5). The deck now asks the structural question outright: “How do we minimize the Bayes risk with squared error loss? (The posterior expectation!)”—and, just above it, plants the hint: “Is this the same structure?” It is. In the Bayesian computation the parameter is a random variable, the data are the predictor, and § 2.8 has already done all the work.

Definition 2.9.1

The posterior distribution

Let the parameter have prior density $\pi(\theta)$ (1.6.1) and the data have conditional density $f(x \mid \theta)$ given θ .⁶ The *posterior distribution* of θ given the observed data $X = x$ is the conditional distribution of θ given $X = x$, with density

$$\pi(\theta \mid x) = \frac{\pi(\theta) f(x \mid \theta)}{\int_{\Theta} \pi(t) f(x \mid t) dt}.$$

⁶Notation shift, deliberate: in Bayesian computations θ is a random variable, so the model density $f(x; \theta)$ is now honestly a *conditional* density and is written $f(x \mid \theta)$ —the same numbers, upgraded punctuation (cf. 1.6.2).

Remark 2.9.2*Reading the posterior*

The numerator is *prior times likelihood*: the data enter the update through $\text{lik}(\theta) = f(x \mid \theta)$ and through nothing else—2.4.1’s object turns out to be the entire interface between data and belief. The denominator does not involve θ : it is one number, the normalizer that makes the slice a probability distribution (1.4.14 again), which is why working Bayesians write $\pi(\theta \mid x) \propto \pi(\theta) f(x \mid \theta)$ and match kernels, as every example below will. And the posterior is the honest owner of sentences the likelihood could not license (2.4.2): given a prior, “ θ is probably near 0.4” is now a statement with a probability attached—computed from $\pi(\cdot \mid x)$.

Theorem 2.9.3

The posterior mean minimizes the Bayes risk

Under squared-error loss, and provided the prior gives θ a finite second moment ($E\{\theta^2\} < \infty$), among all estimators $\delta(X)$ the Bayes risk (1.6.2)

$$r(\delta) = E\{(\delta(X) - \theta)^2\} \quad (\theta \text{ and } X \text{ jointly distributed: prior, then model})$$

is minimized by the *posterior mean*,

$$\delta(x) = E\{\theta \mid X = x\}.$$

Proof 2.9.3

Proof of the Posterior-Mean Rule by Recognizing the Same Structure.

- (1) Under the joint distribution of (θ, X) —prior for θ , model for X given θ —the Bayes risk of a rule δ is exactly a mean squared *prediction* error: the quantity being guessed, θ , is a random variable, and the guess, $\delta(X)$, is a function of the observed one. This is the deck’s “same structure”: 2.8.2 with (X, Y) played by (X, θ) .
- (2) Apply 2.8.2’s argument. Iterate the expectation through the data (1.4.16):

$$r(\delta) = E\left[E\{(\delta(X) - \theta)^2 \mid X\}\right].$$

- (3) Fix the slice $X = x$: inside it $\delta(x)$ is a constant, and the conditional distribution of θ given $X = x$ is the *posterior* (2.9.1). By 2.8.1 applied to that distribution, the inner expectation

$$E\{(\theta - \delta(x))^2 \mid X = x\} = \text{Var}(\theta \mid X = x) + (E\{\theta \mid X = x\} - \delta(x))^2$$

is minimized, slice by slice, at $\delta(x) = E\{\theta \mid X = x\}$.

- (4) Minimizing inside every slice minimizes the average over slices, so the posterior mean minimizes r —the argument that ran once, in miniature and for the uniform prior, in 1.A.5(5). ■

Uses 1.4.16, 2.8.1, 2.8.2.

So the Bayes estimate is: *update, then average*. What remains is to see the update in closed form, and the deck names the three classical cases—“Normal, normal; Binomial, Beta; Poisson, Gamma”—each a *conjugate* pair, in which the posterior lands back in the prior’s own family and the posterior mean is a weighted compromise.

Example 2.9.4

Normal–normal

Prior $\theta \sim N(m, \tau^2)$; one observation $X \mid \theta \sim N(\theta, \sigma^2)$ with σ^2 known (a sample of size n enters as $\bar{X}$ with σ^2/n in place of σ^2). Matching kernels—the full completing-the-square, every step, is §2.A.4—the posterior is again normal,

$$\theta \mid X = x \sim N(m^*, v), \quad \frac{1}{v} = \frac{1}{\tau^2} + \frac{1}{\sigma^2}, \quad m^* = v \left(\frac{m}{\tau^2} + \frac{x}{\sigma^2} \right) = \frac{\sigma^2 m + \tau^2 x}{\sigma^2 + \tau^2}.$$

Read it in *precisions* (inverse variances, the information’s units, 1.9.8): precisions add, and the posterior mean weighs prior guess and data by their precisions. Equivalently, the Bayes estimate

$$E\{\theta \mid X = x\} = m + \frac{\tau^2}{\sigma^2 + \tau^2} (x - m)$$

moves from the prior mean toward the data, covering only the fraction $\tau^2/(\sigma^2 + \tau^2)$ of the distance: it *shrinks* the observation toward m . Vague prior ($\tau^2 \rightarrow \infty$): the estimate is x , the data win. Precise prior ($\tau^2 \rightarrow 0$): the estimate is m , the data are ignored. Everything in between is a negotiation at rates set by the two variances.

Example 2.9.5

Binomial–beta

Prior $\theta \sim \text{Beta}(\alpha, \beta)$ (1.8.5); data $X \mid \theta \sim \text{Binomial}(n, \theta)$. Match kernels in θ :

$$\pi(\theta \mid x) \propto \underbrace{\theta^{\alpha-1}(1-\theta)^{\beta-1}}_{\text{prior}} \cdot \underbrace{\theta^x(1-\theta)^{n-x}}_{\text{likelihood}} = \theta^{(\alpha+x)-1} (1-\theta)^{(\beta+n-x)-1},$$

the $\text{Beta}(\alpha + x, \beta + n - x)$ kernel: successes reinforce α , failures reinforce β . By the Beta mean formula (1.8.5), the Bayes estimate is

$$E\{\theta \mid X = x\} = \frac{\alpha + x}{\alpha + \beta + n} = \frac{(\alpha + \beta) \cdot \frac{\alpha}{\alpha + \beta} + n \cdot \frac{x}{n}}{(\alpha + \beta) + n}:$$

a weighted average of the prior mean $\alpha/(\alpha + \beta)$ and the observed rate x/n , with weights $\alpha + \beta$ and n —the prior is worth $\alpha + \beta$ observations. And with $\alpha = \beta = 1$ (the uniform prior), the estimate is $(1 + x)/(2 + n)$: Unit 1’s shaded estimator, whose origin 1.A.5(4) computed ad hoc, now a two-

line corollary of the general theorem—the promise “where $(1 + X)/(2 + n)$ comes from,” kept and generalized.

Example 2.9.6

Poisson–gamma

Prior $\theta \sim \text{Gamma}(a, b)$, density proportional to $\theta^{a-1}e^{-b\theta}$ on $(0, \infty)$ (mean a/b); data $X_1, \dots, X_n \mid \theta$ iid $\text{Poisson}(\theta)$, likelihood proportional to $\theta^{\sum_i x_i} e^{-n\theta}$. The product is the $\text{Gamma}(a + \sum_i x_i, b + n)$ kernel, so

$$\mathbb{E}\{\theta \mid \sum_i X_i = s\} = \frac{a+s}{b+n} = \frac{b \cdot \frac{a}{b} + n \cdot \bar{x}}{b+n} :$$

prior mean weighted b , sample mean weighted n . The prior acts like b phantom observations totaling a events.

Remark 2.9.7

One anatomy, three posteriors

All three estimates are the same sentence:

$$\text{posterior mean} = \frac{(\text{prior worth}) \times (\text{prior guess}) + (\text{data worth}) \times (\text{data guess})}{(\text{prior worth}) + (\text{data worth})},$$

with “worth” measured in observation-equivalents (or precisions, 2.9.4). As data accumulate the data’s worth grows without bound and the Bayes estimate slides toward the data’s guess; with little data the prior holds it back. That systematic holding-back has a name, and the deck’s next slide says it.

2.10 Shrinkage and Penalization

“By the way,” the seventeenth slide begins—and then quietly names the modern half of the subject.

Concept 2.10.1

Bayes as penalized likelihood, Bayes as shrinkage

“Notice how Bayes methods are sort of like penalized likelihood methods; ‘shrinkage’ estimators.” Both readings are visible in objects already on the table. *Penalty*: by 2.9.1, maximizing the posterior means maximizing

$$\log \pi(\theta \mid x) = \ell(\theta) + \log \pi(\theta) + \text{const} :$$

the log-likelihood *plus a penalty term* that docks points for straying where the prior is thin. A normal prior contributes $\log \pi(\theta) = -(\theta - m)^2/(2\tau^2) + \text{const}$, a quadratic penalty on distance from m —the ridge-regression form, with $1/\tau^2$ as the tuning knob. *Shrinkage*: the posterior-mean formulas of § 2.9 do not merely resemble shrinking—they *are* affine maps pulling the data’s guess partway toward the prior’s, by factors like $\tau^2/(\sigma^2 + \tau^2)$ and $n/(\alpha + \beta + n)$. The deck’s hedge (“sort of like”) is apt: the posterior *mean* and the posterior *maximizer* are different summaries of the same posterior—they coincide in the normal–normal case and differ in general—but the family resemblance among Bayes, penalization, and shrinkage is structural, not cosmetic.

Remark 2.10.2

The deck’s big-data point

The slide closes: “At the heart of first-order big data: you cannot use the data to know what to shrink towards; but you can use the data to know how much to shrink towards it.” Unpacked: the shrinkage *target* (m ; the prior mean) plays the role of outside knowledge—estimating it from the same data that will then be shrunk toward it is circular, at least to first order. But the shrinkage *amount*—the weight $\tau^2/(\sigma^2 + \tau^2)$ —is a different kind of quantity: with many parallel problems (many arms, many genes, many small areas), the spread of the observed estimates around the target reveals τ^2 , and the data can tune their own shrinkage. Estimating the prior’s parameters from the ensemble is called *empirical Bayes*, and the discovery that such procedures can beat the obvious unbiased one at its own game (Stein’s phenomenon) is one of the subject’s genuine surprises—here, only a signpost at the edge of the course.

2.11 Looking Ahead: Sufficiency

Concept 2.11.1

“If we are done by Thursday”

The deck’s final slide is a conditional promise, and a map of the next idea: *sufficient statistics*—summaries of the data that lose nothing about the parameter (note how every example this week funneled the whole sample through $\bar{X}$ or a total; Fisher’s “reduction of data,” the epigraph, is this program); *conditioning on sufficient statistics*, whose payoffs the slide indexes only as “Unbiasedness” and “Relationship to variance”—the mathematics to come: conditioning preserves unbiasedness and never increases variance, so any unbiased estimator can be upgraded by averaging it over the sufficient summary; and *complete* sufficient statistics, whose “implications for uniqueness” deliver the unique uniformly optimal unbiased estimator that 2.1.7 advertised. The deck defers the mathematics, and so do these notes: the objects above are next week’s, named here so the week’s arc—floor, constructor, precision, prior—is visible as one story.

Appendix

Supplementary material—additional proofs, context, and other treatments; not part of the lecture proper.

The deck names its computations and leaves them to the board and to us; the longer ones are worked here, every step shown, each keyed to the object it supports. §2.A.1 settles the sample variance’s bias (2.1.6, 2.6.3); §2.A.2 prices nuisance parameters (2.2.5); §2.A.3 computes the exponential rate’s bias and the variance of its repair (2.5.4); §2.A.4 completes the square behind the normal–normal posterior (2.9.4); and §2.A.5 runs Fieller’s quadratic through its cases (2.7.7).

Supplement 2.A.1

The sample variance’s bias, and Bessel’s correction

For $X_1, \dots, X_n$ iid with mean μ and variance σ^2 ,

$$\mathbb{E}\left\{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2\right\} = \frac{n-1}{n} \sigma^2, \quad \text{so} \quad \mathbb{E}\left\{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2\right\} = \sigma^2.$$

Proof 2.A.1

Proof of Bessel’s Correction by Adding and Subtracting the True Mean.

- (1) Route each deviation through μ : $X_i - \bar{X} = (X_i - \mu) - (\bar{X} - \mu)$, square, and sum:

$$\sum_i (X_i - \bar{X})^2 = \sum_i (X_i - \mu)^2 - 2(\bar{X} - \mu) \sum_i (X_i - \mu) + n(\bar{X} - \mu)^2.$$

- (2) The middle sum telescopes into the average: $\sum_i (X_i - \mu) = n(\bar{X} - \mu)$, so the middle term is $-2n(\bar{X} - \mu)^2$ and

$$\sum_i (X_i - \bar{X})^2 = \sum_i (X_i - \mu)^2 - n(\bar{X} - \mu)^2.$$

(Measuring from the sample’s own center *always* removes exactly $n(\bar{X} - \mu)^2$ worth of scatter—the algebraic content of “ $\bar{X}$ has chased the data,” 2.1.6(c).)

- (3) Take expectations: $\mathbb{E}\{(X_i - \mu)^2\} = \sigma^2$ for each i , and $\mathbb{E}\{(\bar{X} - \mu)^2\} = \text{Var}(\bar{X}) = \sigma^2/n$, so

$$\mathbb{E}\left\{\sum_i (X_i - \bar{X})^2\right\} = n\sigma^2 - n \cdot \frac{\sigma^2}{n} = (n-1)\sigma^2.$$

- (4) Divide by n for the plug-in’s expectation, $(n-1)\sigma^2/n$; divide by $n-1$ instead and the expectation is exactly σ^2 . ■

Supplement 2.A.2

Nuisance parameters cost information: the 2×2 case

Let $I = \begin{pmatrix} I_{11} & I_{12} \\ I_{12} & I_{22} \end{pmatrix}$ be an information matrix (symmetric, positive definite). Then

$$(I^{-1})_{11} = \frac{I_{22}}{I_{11}I_{22} - I_{12}^2} \geq \frac{1}{I_{11}}, \quad \text{with equality if and only if } I_{12} = 0.$$

Proof 2.A.2

Proof of the Nuisance Cost by the Explicit Inverse.

- (1) The 2×2 inverse: with $\det I = I_{11}I_{22} - I_{12}^2 > 0$ (positive definiteness),

$$I^{-1} = \frac{1}{I_{11}I_{22} - I_{12}^2} \begin{pmatrix} I_{22} & -I_{12} \\ -I_{12} & I_{11} \end{pmatrix}, \quad \text{so} \quad (I^{-1})_{11} = \frac{I_{22}}{I_{11}I_{22} - I_{12}^2}.$$

- (2) Compare with $1/I_{11}$ by subtracting, over the common positive denominator:

$$(I^{-1})_{11} - \frac{1}{I_{11}} = \frac{I_{11}I_{22} - (I_{11}I_{22} - I_{12}^2)}{I_{11}(I_{11}I_{22} - I_{12}^2)} = \frac{I_{12}^2}{I_{11}(I_{11}I_{22} - I_{12}^2)} \geq 0,$$

every factor in the denominator positive.

- (3) The difference vanishes exactly when $I_{12} = 0$: the orthogonal case. Correlated scores mean the nuisance coordinate's uncertainty leaks into yours; the leak is the ratio displayed. ■

Supplement 2.A.3

The exponential rate: bias of $1/\bar{X}$, and the repaired estimator's variance

Let $X_1, \dots, X_n$ be iid Exponential with rate λ and let $S = \sum_i X_i$. Then for $n \geq 2$,

$$\mathbb{E}^\lambda \left\{ \frac{n}{S} \right\} = \frac{n\lambda}{n-1},$$

so the MLE $\hat{\lambda} = 1/\bar{X} = n/S$ is biased upward; and the unbiased repair $(n-1)/S$ has, for $n \geq 3$,

$$\text{Var}^\lambda \left(\frac{n-1}{S} \right) = \frac{\lambda^2}{n-2} > \frac{\lambda^2}{n} = \frac{1}{I_n(\lambda)}.$$

Proof 2.A.3

Proof of the Bias and the Repaired Variance via the Gamma Integral.

- (1) The total of n iid Exponential(λ) variables has the Gamma(n, λ) density $f_S(s) = \lambda^n s^{n-1} e^{-\lambda s} / \Gamma(n)$, $s > 0$ (a standard fact; it can be built by induction on n).

- (2) Negative first moment, by matching the integrand back to a Gamma density of lower order:

$$\mathbb{E}\left\{\frac{1}{S}\right\} = \int_0^\infty \frac{1}{s} \cdot \frac{\lambda^n s^{n-1} e^{-\lambda s}}{\Gamma(n)} ds = \frac{\lambda^n}{\Gamma(n)} \int_0^\infty s^{n-2} e^{-\lambda s} ds,$$

and the surviving integral is the Gamma integral one order down, $\Gamma(n-1)/\lambda^{n-1}$, so

$$\mathbb{E}\left\{\frac{1}{S}\right\} = \frac{\lambda^n}{\Gamma(n)} \cdot \frac{\Gamma(n-1)}{\lambda^{n-1}} = \frac{\lambda}{n-1},$$

using $\Gamma(n) = (n-1)\Gamma(n-1)$ and requiring $n \geq 2$ for the integral to converge. Multiply by n : $\mathbb{E}\{n/S\} = n\lambda/(n-1)$.

- (3) Negative second moment, the same maneuver one order down:

$$\mathbb{E}\left\{\frac{1}{S^2}\right\} = \frac{\lambda^n}{\Gamma(n)} \cdot \frac{\Gamma(n-2)}{\lambda^{n-2}} = \frac{\lambda^2}{(n-1)(n-2)} \quad (n \geq 3).$$

- (4) Hence for the estimator $(n-1)/S$ (unbiased by step (2), since $\mathbb{E}\{(n-1)/S\} = (n-1) \cdot \lambda/(n-1) = \lambda$):

$$\text{Var}\left(\frac{n-1}{S}\right) = (n-1)^2 \left[\frac{\lambda^2}{(n-1)(n-2)} - \frac{\lambda^2}{(n-1)^2} \right] = \lambda^2 \left[\frac{n-1}{n-2} - 1 \right] = \frac{\lambda^2}{n-2},$$

strictly above the floor λ^2/n : in this parameterization no estimator in sight attains the bound—“a not sharp bound” indeed (1.9.14). ■

Uses 2.5.4.

Supplement 2.A.4

The normal–normal posterior, every step

With prior $\theta \sim N(m, \tau^2)$ and one observation $X \mid \theta \sim N(\theta, \sigma^2)$, the posterior is $N(m^*, v)$ with

$$v = \left(\frac{1}{\tau^2} + \frac{1}{\sigma^2} \right)^{-1}, \quad m^* = v \left(\frac{m}{\tau^2} + \frac{x}{\sigma^2} \right) = \frac{\sigma^2 m + \tau^2 x}{\sigma^2 + \tau^2}.$$

Proof 2.A.4

Proof of the Posterior by Completing the Square in θ .

- (1) Multiply prior and likelihood and keep only θ -dependent factors:

$$\pi(\theta \mid x) \propto \exp\left(-\frac{(\theta - m)^2}{2\tau^2}\right) \exp\left(-\frac{(x - \theta)^2}{2\sigma^2}\right),$$

so, combining the exponents,

$$\pi(\theta | x) \propto \exp\left(-\frac{1}{2} \left[\frac{(\theta - m)^2}{\tau^2} + \frac{(\theta - x)^2}{\sigma^2} \right]\right).$$

(2) Expand the bracket and collect powers of θ :

$$\frac{(\theta - m)^2}{\tau^2} + \frac{(\theta - x)^2}{\sigma^2} = \theta^2 \left(\frac{1}{\tau^2} + \frac{1}{\sigma^2} \right) - 2\theta \left(\frac{m}{\tau^2} + \frac{x}{\sigma^2} \right) + \left(\frac{m^2}{\tau^2} + \frac{x^2}{\sigma^2} \right).$$

(3) Write $\frac{1}{v} = \frac{1}{\tau^2} + \frac{1}{\sigma^2}$ and $m^* = v \left(\frac{m}{\tau^2} + \frac{x}{\sigma^2} \right)$, and complete the square:

$$\theta^2 \cdot \frac{1}{v} - 2\theta \cdot \frac{m^*}{v} = \frac{\theta^2 - 2\theta m^*}{v} = \frac{(\theta - m^*)^2}{v} - \frac{(m^*)^2}{v},$$

and the terms free of θ join the normalizer.

(4) The θ -dependence is exactly $\exp\{-(\theta - m^*)^2/(2v)\}$: the $N(m^*, v)$ kernel. Since the posterior is a probability distribution (2.9.1), it *is* $N(m^*, v)$. Clearing v from m^* 's definition—multiply numerator and denominator by $\sigma^2\tau^2$:

$$m^* = \frac{\frac{m}{\tau^2} + \frac{x}{\sigma^2}}{\frac{1}{\tau^2} + \frac{1}{\sigma^2}} = \frac{\sigma^2 m + \tau^2 x}{\sigma^2 + \tau^2}.$$

■

Uses 2.9.1.

Supplement 2.A.5

Fieller's quadratic, case by case

The set of 2.7.7 is $S = \{\rho : g(\rho) \leq 0\}$ where, writing $c = z_{\alpha/2}^2 \sigma^2$,

$$g(\rho) = \left(\bar{Y}^2 - \frac{c}{m} \right) \rho^2 - 2\bar{X}\bar{Y}\rho + \left(\bar{X}^2 - \frac{c}{n} \right),$$

and its shape is decided by the leading coefficient and the roots.

Proof 2.A.5

Proof of the Case Analysis by Expanding the Inversion Inequality.

(1) Square out the defining inequality $(\bar{X} - \rho\bar{Y})^2 \leq c \left(\frac{1}{n} + \frac{\rho^2}{m} \right)$ and move everything to the left:

$$\bar{X}^2 - 2\bar{X}\bar{Y}\rho + \bar{Y}^2\rho^2 - \frac{c}{n} - \frac{c}{m}\rho^2 \leq 0,$$

which is $g(\rho) \leq 0$ with g as displayed.

- (2) The discriminant of g , after expanding and cancelling the $\bar{X}^2\bar{Y}^2$ terms:

$$4\bar{X}^2\bar{Y}^2 - 4\left(\bar{Y}^2 - \frac{c}{m}\right)\left(\bar{X}^2 - \frac{c}{n}\right) = 4c\left(\frac{\bar{Y}^2}{n} + \frac{\bar{X}^2}{m} - \frac{c}{nm}\right).$$

- (3) In the case of a positive leading coefficient ($\bar{Y}^2 > c/m$: the denominator mean is convincingly nonzero), the parabola opens upward and the discriminant is strictly positive: since $\bar{Y}^2 > c/m$, its parenthesis strictly exceeds $\frac{c/m}{n} + \frac{\bar{X}^2}{m} - \frac{c}{nm} = \frac{\bar{X}^2}{m} \geq 0$. Two real roots; S is the genuine interval between them.
- (4) In the case of a negative leading coefficient ($\bar{Y}^2 < c/m$: the data cannot rule out $\mu_2 = 0$), the parabola opens downward. If the discriminant is positive, $g \leq 0$ outside the roots: S is two half-lines running to $\pm\infty$. If it is negative, $g < 0$ everywhere: $S = \mathbb{R}$ —the data place no restriction on the ratio at all.
- (5) Coverage is untouched by the case analysis: whatever its shape, S contains the true ρ exactly when $|Z(\rho)| \leq z_{\alpha/2}$, an event of probability $1 - \alpha$ (2.7.4). When the denominator might be zero the ratio is genuinely ill-determined, and an honest $1 - \alpha$ report says so by refusing to be an interval. ■

Uses 2.7.7.

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